Static D2NN Loss Function Sweep

Practical Limits & Optimal Loss Strategy

D2NN Lab | 2026-04-03

FSO Beam Cleanup $C_n^2=5\times10^{-14}$, L=1km, λ=1.55µm

Experiment Overview

10 loss strategies across 3 sweeps, 2 datasets

Sweep	Data	Strategies	Status
0401	dn100um (Lanczos)	PIB, Strehl, IO, CO+PIB	Complete
0402	pitch_rescale	Abs Bucket (F), TP+PIB (B), Multiplane (C)	Complete
0403	pitch_rescale	CO+hardTP, AbsBkt+CO, PIB+hardTP	Complete

10
Loss Strategies

5000
Training Pairs

30
Epochs Each

~3h
Per Strategy (A100)

`5-layer D2NN | spacing=10mm | f=4.5mm | N=1024 | dx=2µm | pad=2`

Normalized PIB Lies

High PIB does NOT mean high received power

Strategy	PIB@10µm	Throughput	Abs Power vs Turb	Verdict
PIB Only (0401)	90.1%	50.7%	58.7%	LOSS
Strehl Only (0401)	81.8%	57.5%	60.4%	LOSS
Abs Bucket F (0402)	81.4%	98.6%	105.8%	GAIN
TP+PIB B (0402)	81.6%	98.4%	105.8%	GAIN
Multiplane C (0402)	84.2%	39.5%	44.2%	LOSS
CO+hardTP (0403)	74.0%	99.3%	97.0%	~NEUTRAL

Key Insight: PIB Only achieved 90% PIB but absolute received power was only 58.7% of turbulent baseline. **Throughput is the real metric.**

Why Does Throughput Drop?

Spatial Filtering Mechanism

1. **Phase Mask Pattern**
   D2NN learns high-frequency phase gratings to redirect energy

2. **Angular Vignetting**
   Diffracted light at large angles exceeds finite aperture ($dx=2\mu m \gg \lambda/2=775nm$)

3. **Aperture Spatial Filter**
   Energy scattered beyond computational domain is lost at each layer

4. **Energy Lost**
   Turbulent beams lose MORE energy (50%) than vacuum (20-30%)

D2NN acts as aperture-based spatial filter — diffracts energy beyond collection aperture

Received Power Gap Decomposition

	Total Power	Bucket@10µm	PIB
Vacuum	290.18	277.17	95.5%
Turbulent	278.67	217.95	78.1%
Gap	**11.51 (4.0%)**	**59.22 (21.4%)**	**17.4%p**

Physical Parameters

WFE = 442.9 nm (2.87λ)

Maréchal limit = λ/14 = 111nm

Strehl = $\exp(-\sigma^2)$ = 0.04

Gap origin:

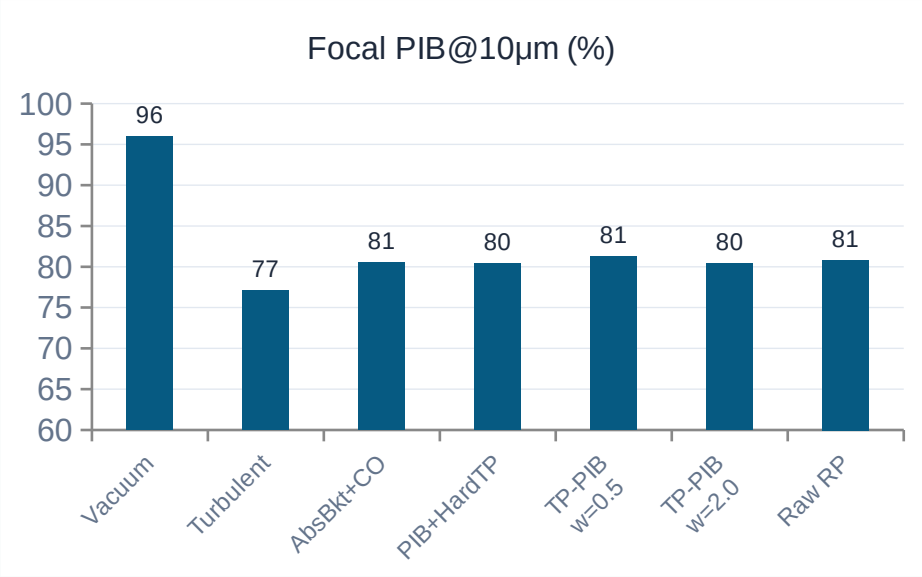
- WFE spreading: 81.8%

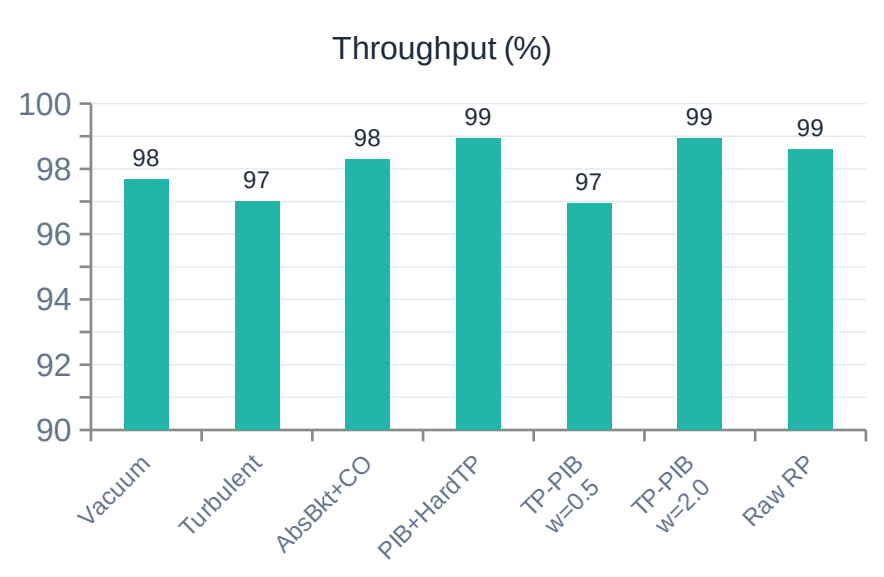
$$\text{Bucket Power} \approx \text{TP} \times \exp(-\sigma^2_\text{WFE}) \times \text{P_diff.limited}$$

Structural Limit: Each sample has random WFE from different turbulence realization. Fixed masks cannot correct random per-sample error → fundamental constraint.

0403 Combined 6-Strategy Results

Focal-plane PIB@10μm with pitch_rescale data, $Cn^2=5\times10^{-14}$

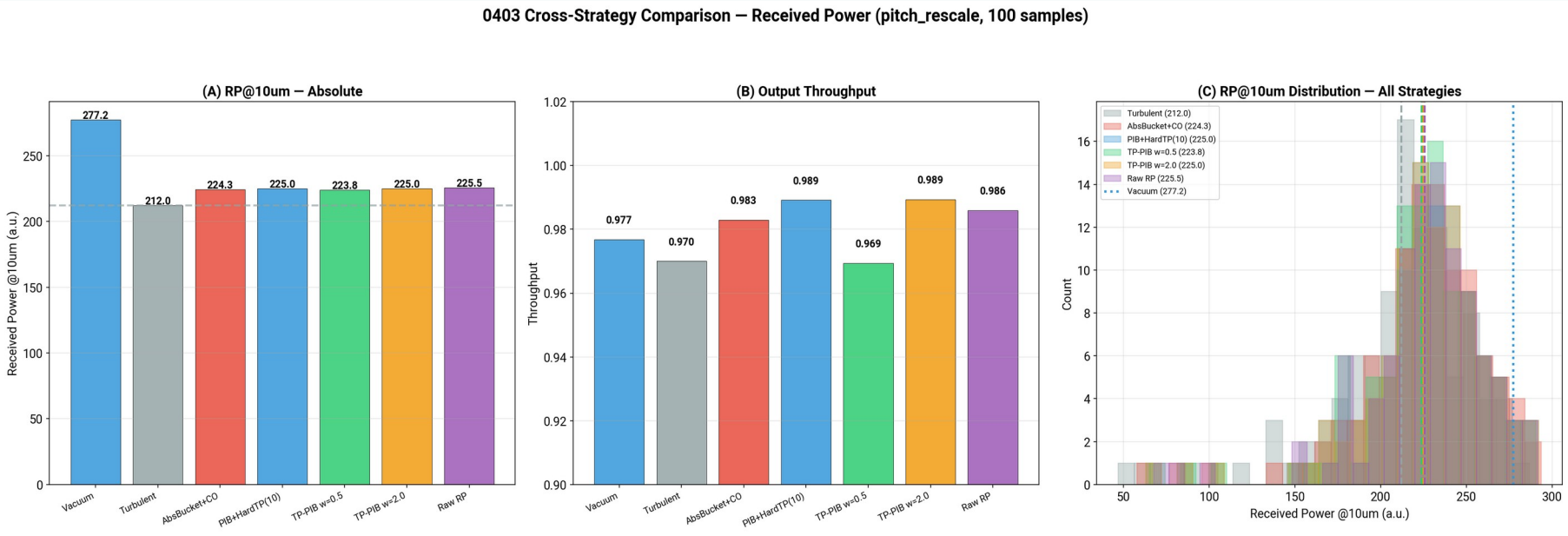




Best strategy: TP-PIB w=0.5 achieves highest PIB (81.3%) while maintaining 96.9% throughput. All 0403 strategies preserve throughput >96%.

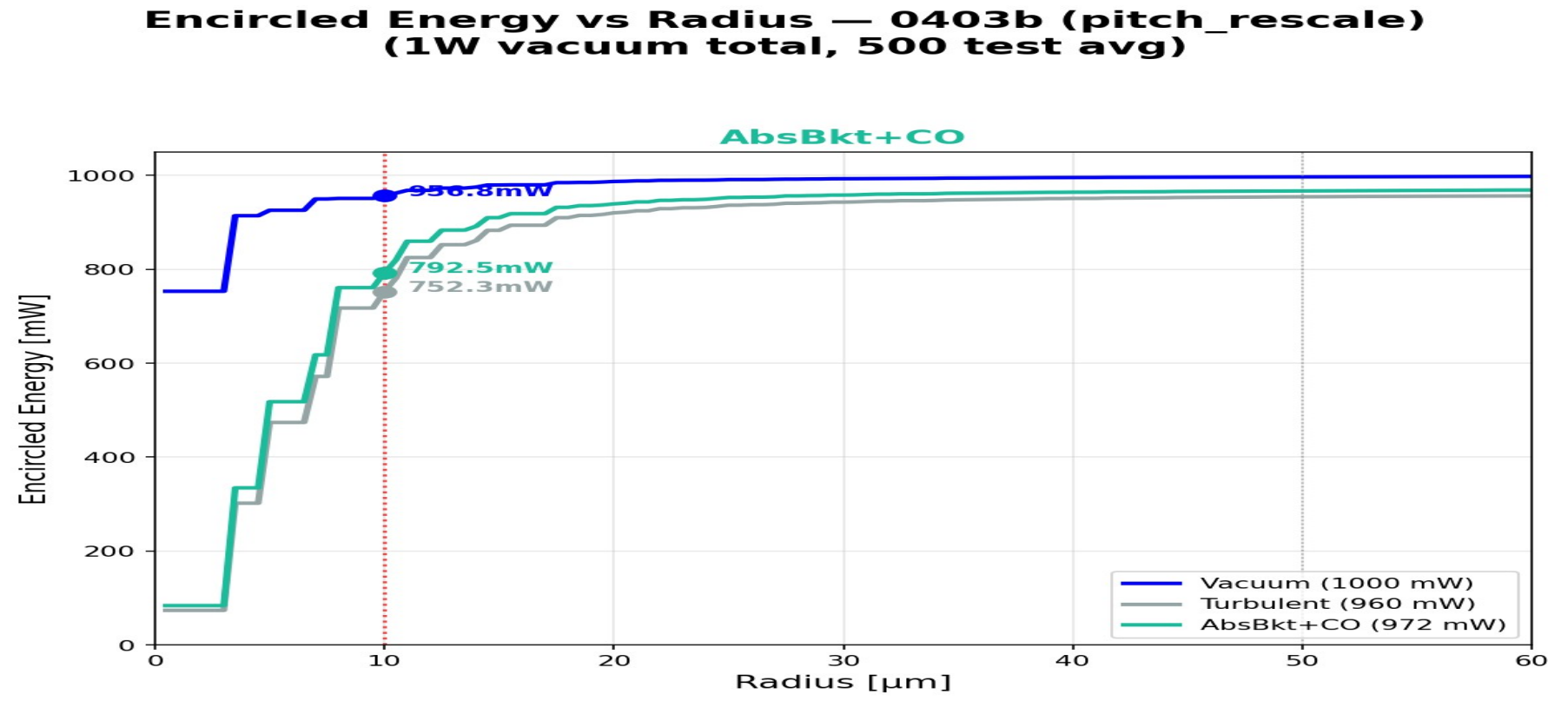
Absolute Received Power Distribution

Turbulent (gray) vs D2NN (color) — 500 test samples, bucket @10µm



+4.0%	+3.3%	+5.2%	+3.3%	+3.8%
AbsBkt+CO	PIB+HardTP	TP-PIB w=0.5	TP-PIB w=2.0	Raw RP

Encircled Energy Analysis



PIB computed at 4 radii: 5µm, 10µm, 25µm, 50µm | All strategies converge at r=50µm

CO Loss: High Throughput, Low PIB

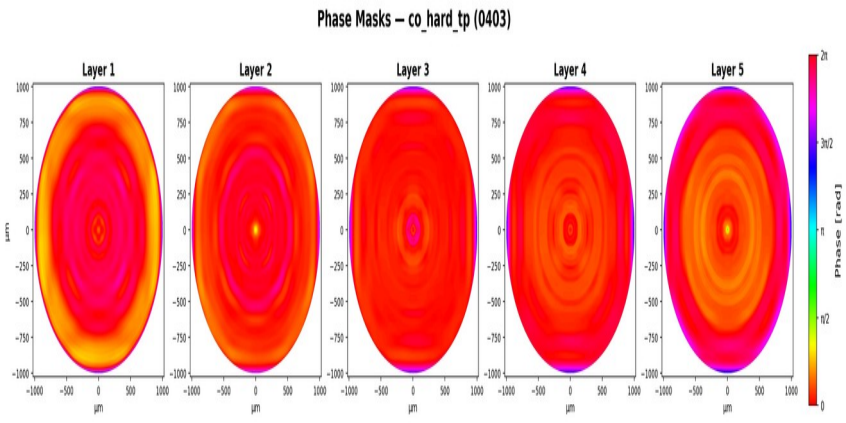
CO+hardTP (0403) — Complex Overlap + Quadratic TP Penalty (λ=10)

99.3% — **Throughput**
Best across all strategies

74.0% — **PIB@10µm**
Below turbulent baseline (76.1%)

97.0% — **Abs Power**
3% loss vs turbulent

0.49 — **Strehl**
Far below diffraction limit



Phase masks — nearly flat (minimal spatial filtering)

Analysis: CO optimizes entire field matching, not bucket concentration. No gradient drives energy into the bucket. Highest TP but worst PIB — confirms CO alone is insufficient for FSO beam cleanup.

Deep Fade & Scintillation Analysis

Scintillation Statistics (dB, vacuum mean = 0 dB)

	Mean	Std	Min	1%ile	5%ile
Vacuum	0.00	0.00	0.00	0.00	0.00
Turbulent	-1.12	0.86	**-7.73**	-3.72	-2.45
D2NN (F)	-0.85	0.74	**-6.36**	-3.25	-2.00

Deep Fade Worst Cases

Rank	Turb	D2NN	Gain
1	-7.73 dB	-6.36 dB	**+1.37 dB**
2	-6.38 dB	-5.20 dB	**+1.18 dB**
3	-6.28 dB	-5.05 dB	**+1.24 dB**
8	-3.68 dB	-2.66 dB	**+1.02 dB**

Fade Probability

Threshold	Turbulent	D2NN (F)
< -3 dB	2.8%	**1.8%**
< -5 dB	0.6%	0.6%
< -7 dB	0.2%	**0.0%**

D2NN is more effective in deep fades (+1.37 dB worst case) than average (+0.24 dB). **Static D2NN acts as a "floor raiser" for worst-case link budget.**

Zernike Mode Analysis

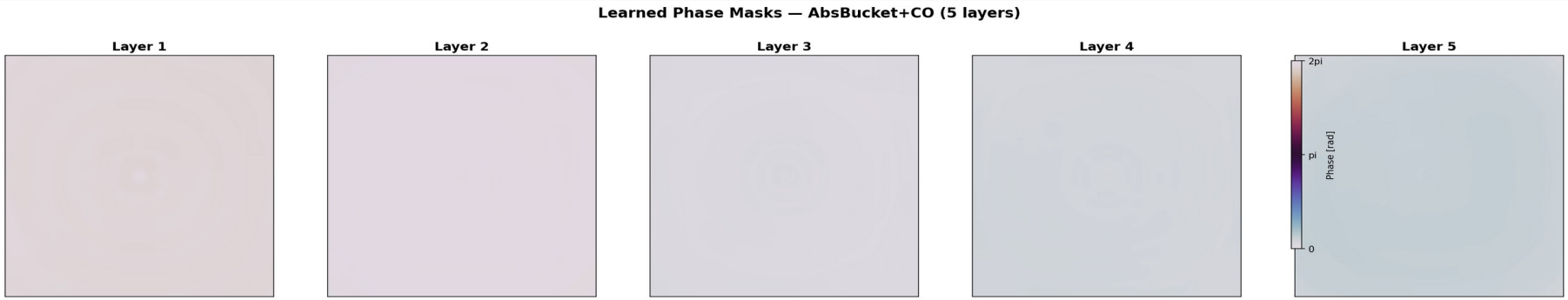
Wavefront correction capability by loss strategy

Strategy	Higher-order Δ	Tip/Tilt Δ	Throughput
F/B (Abs Bucket)	±1 nm	±0.2 nm	98%
Multiplane C	-28 nm	+1.3 nm	40%
PIB Only	-28 nm	+16 nm	51%

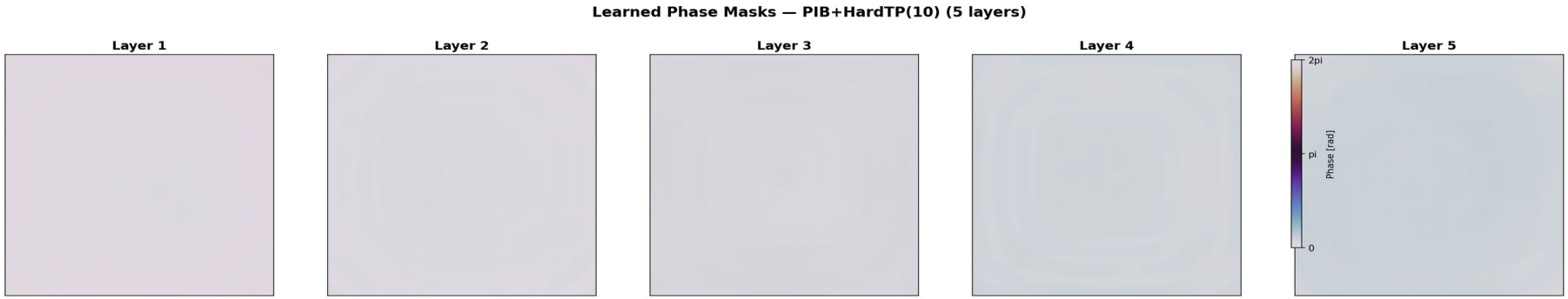
PIB ↔ Throughput : Structural Tradeoff

- When TP is preserved (F/B), Zernike delta ≈ 0 — no wavefront correction
- When TP drops (PIB Only), higher-order corrected -28 nm but tip/tilt added +16 nm
- Static mask trades one aberration type for another via mode conversion
- This confirms: Fixed masks cannot correct random wavefront errors

Phase Mask Comparison



AbsBkt+CO — nearly flat masks (TP=98.3%)



PIB+HardTP — also flat masks due to strong TP penalty (TP=98.9%)

All 0403 strategies learn conservative masks — TP penalty prevents aggressive spatial filtering

Optimal Loss Function

Absolute Bucket Loss — direct fiber-coupling optimization

$$L_{abs} = 1 - \Sigma_{r\leq 10\mu m} |E_{pred}^{focal}|^2 / \Sigma_{r\leq 10\mu m} |E_{vac}^{focal}|^2$$

1. **Throughput-aware**
   Normalizing by vacuum bucket power captures both TP and PIB

2. **Gradient-aligned**
   Every photon directed into bucket reduces loss directly

3. **Physics-faithful**
   Bucket Power ∝ TP × Strehl × P_diff → loss aligns with all three

Conclusions

1. Throughput is the real performance metric. Normalized PIB is misleading.
2. Static D2NN absolute power improvement: +5.8% (+0.24 dB) — Abs Bucket (F) / TP+PIB (B).
3. PIB ↔ Throughput structural tradeoff. Energy concentration via scatter → throughput loss unavoidable.
4. Vacuum PIB (94.2%) is TP-preserved ceiling. WFE=442.9nm (random per-sample) → fixed masks can't correct.
5. Deep fade improvement: +1.37 dB worst case. Static D2NN is a "floor raiser".
6. Absolute bucket loss is the optimal objective. Bucket Power ∝ TP × Strehl × P_diff.

Next: Stronger turbulence ($Cn^2=10^{-13}$) | Switchable static masks | Hybrid AO+D2NN | Dynamic D2NN (Zernike-based)

Physics Verification

Expert review of equations and physical claims (Codex/GPT-5.4 analysis)

Maréchal Strehl = $\exp(-\sigma^2)$	Exact for Gaussian phase stats, not just small-aberration	CORRECT
$L_{abs} = 1 - \Sigma\lvert E_{pred}\rvert^2 / \Sigma\lvert E_{vac}\rvert^2$	Implicitly couples TP+PIB gradients; ideal for fiber	CORRECT
Bucket $\propto$ TP $\times \exp(-\sigma^2) \times$ P_diff	Assumes separable TP/WFE; breaks in strong scintillation	APPROX
Spatial filtering mechanism	Angular vignetting, NOT evanescent ($dx=2\mu m \gg \lambda/2$)	CORRECTED
PIB $\leftrightarrow$ TP structural tradeoff	Real for static masks; not information-theoretic limit	CORRECT
Unitary CO preservation $\lvert\Delta CO\rvert<0.01$	Breaks with finite aperture + multi-layer losses	CORRECT
Zernike tip/tilt addition	Fixed mask adds fixed Zernike; modes affected differently	CORRECT
Deep fade floor effect	Spatial filtering establishes quality floor; expected	CORRECT

Reviewed by Codex (GPT-5.4) | 6/8 fully correct, 1 approximately correct, 1 terminology corrected